THE PRESHEAF OF SECTIONS OF A PULLBACK OF A QUASI-COHERENT SHEAF SATISFIES THE FPQC SHEAF CONDITION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The two same-chapter pointers are located at their \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- The fpqc topology and the sheaf condition for it. Standard in this chapter; the statement refers to the presheaf F^a on Sch/S and to the sheaf condition for the fpqc topology. Supplies the reduction of the sheaf condition to Zariski coverings and faithfully flat morphisms of affine schemes; the verification in both cases is the counted content.
- Descent of modules along a faithfully flat ring map. Same chapter, Lemma labelled lemma-descent-modules (source line 1237, Stacks tag 03OA); invoked at proof lines 1364-1365. Supplies exactness of 0 -> M -> B tensor_A M -> B tensor_A B tensor_A M for A -> B faithfully flat; the translation of that exactness into the sheaf condition is retained in the counted span.

Source reference. Original lemma 03DT: sources/stacks-chow/tags.txt, source lines 4432–4432. Cross-chapter pointer named at proof line 1346 as where the proposition is proved; the author immediately announces 'We indicate the proof here' and the indicated argument in this chapter is retained verbatim in the counted span.

Source reference. Original lemma 03O1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 974–974. Same-chapter pointer invoked at proof lines 1347-1349 for the reduction of the fpqc sheaf condition to Zariski coverings and faithfully flat morphisms of affine schemes; both cases of the verification are retained in the counted span.

Source reference. Original lemma 03OA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1237–1237. Same-chapter pointer invoked at proof lines 1364-1365 for exactness of the Amitsur complex of a faithfully flat ring map; the identification of that exactness with the sheaf condition for F^a is retained in the counted span.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proposition 0.1. *For any quasi-coherent sheaf $\mathcal{F}$ on S the presheaf*

$$\begin{array}{rccc} \mathcal{F}^a : & Sch/S & \to & Ab \\ & (f : T \to S) & \mapsto & \Gamma(T, f^*\mathcal{F}) \end{array}$$

is an $\mathcal{O}$-module which satisfies the sheaf condition for the fpqc topology.

Proof. This is proved in Descent, Lemma 03DT. We indicate the proof here. As established in Lemma 03O1, it is enough to check the sheaf property on Zariski coverings and faithfully flat morphisms of affine schemes. The sheaf property for Zariski coverings is standard scheme theory, since $\Gamma(U, i^*\mathcal{F}) = \mathcal{F}(U)$ when $i : U \hookrightarrow S$ is an open immersion.

For $\{\mathrm{Spec}(B) \to \mathrm{Spec}(A)\}$ with $A \to B$ faithfully flat and $\mathcal{F}|_{\mathrm{Spec}(A)} = \widetilde{M}$ this corresponds to the fact that $M = H^0\left((B/A)_\bullet \otimes_A M\right)$, i.e., that

$$0 \to M \to B \otimes_A M \to B \otimes_A B \otimes_A M$$

is exact by Lemma 03OA. □